

GEOGRAPHY

PHYSICAL GEOGRAPHY



CLIMATOLOGY

MODULE – 24

TROPICAL CYCLONES

- CHARACTERISTICS OF CYCLONES
- RECURVING CYCLONES

TROPICAL CYCLONES



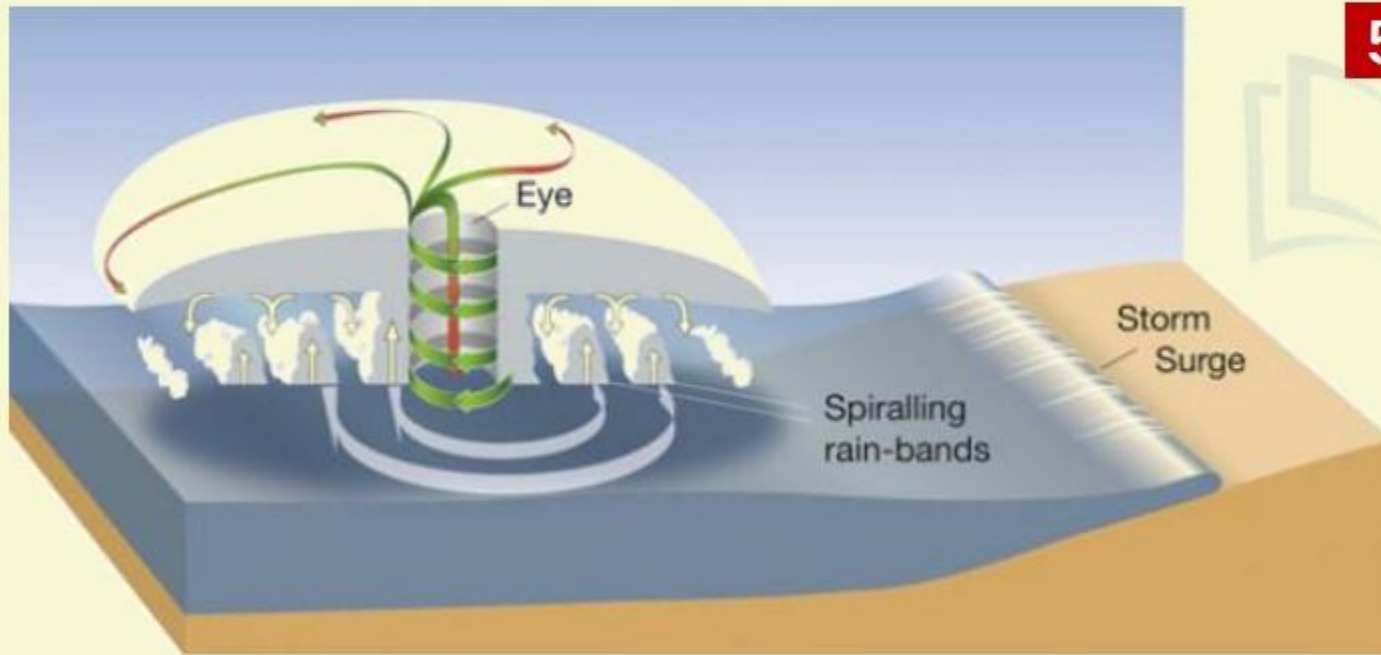
Path of Cyclone Vardah

1 Tropical cyclones are violent storms that **originate over oceans in tropical areas** and move over to the coastal areas bringing about large scale destruction caused by **violent winds, very heavy rainfall and storm surges**.

2 This is one of the most devastating natural calamities.

3 They are known as **Cyclones** in the Indian Ocean, **Hurricanes** in the Atlantic, **Typhoons** in the Western Pacific and South China Sea and **Willy-willies** in the Western Australia.

TROPICAL CYCLONES



4 Tropical cyclones originate and intensify over warm tropical oceans.

5 The conditions favorable for the formation and intensification of tropical storms are:

1. Large sea surface with temperature higher than 27°C .
2. Presence of Coriolis force
3. Minimum variations in the vertical wind speed aka wind shear
4. A pre-existing weak-low-pressure area or low-level-cyclonic circulation
5. Upper air divergence above the sea level system

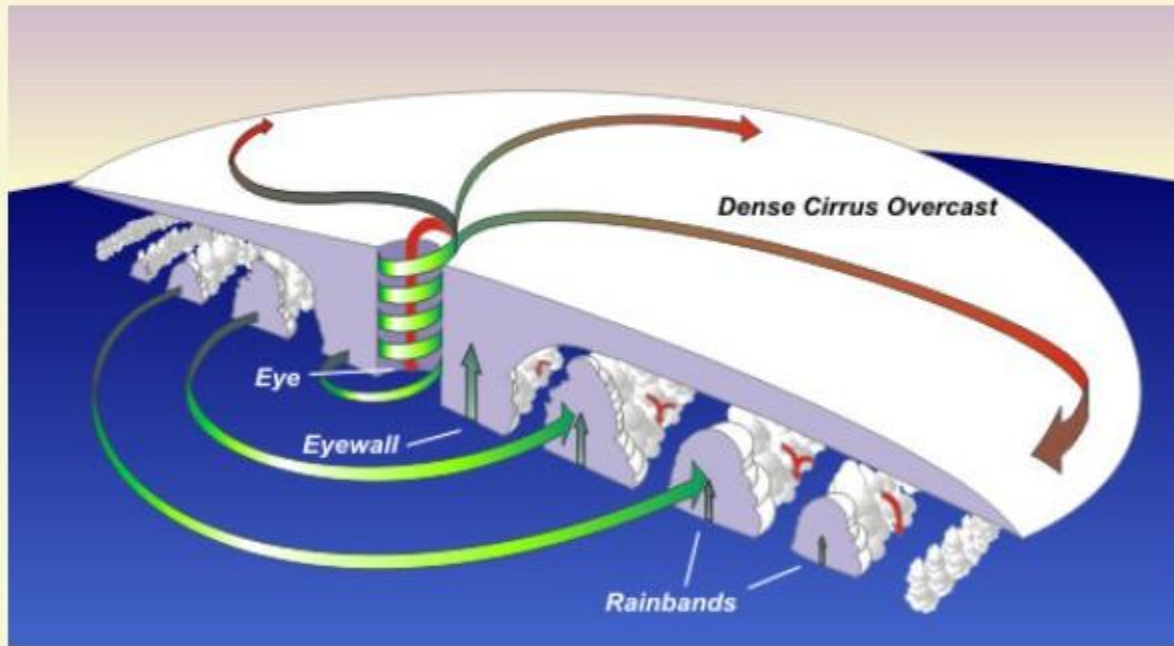
TROPICAL CYCLONES



- 6** The energy that intensifies the storm, comes from the **condensation process** in the **towering cumulonimbus clouds**, surrounding the center of the storm.
- 7** With **continuous supply of moisture** from the sea, the storm is further strengthened.
- 8** On reaching the land the moisture supply is cut off and the storm dissipates.
- 9** The place where the tropical cyclone crosses the coast is called **the landfall of the cyclone**.
- 10** The cyclones, which **cross 20°N latitude** generally **recurve** and they are more **destructive**.

TROPICAL CYCLONES

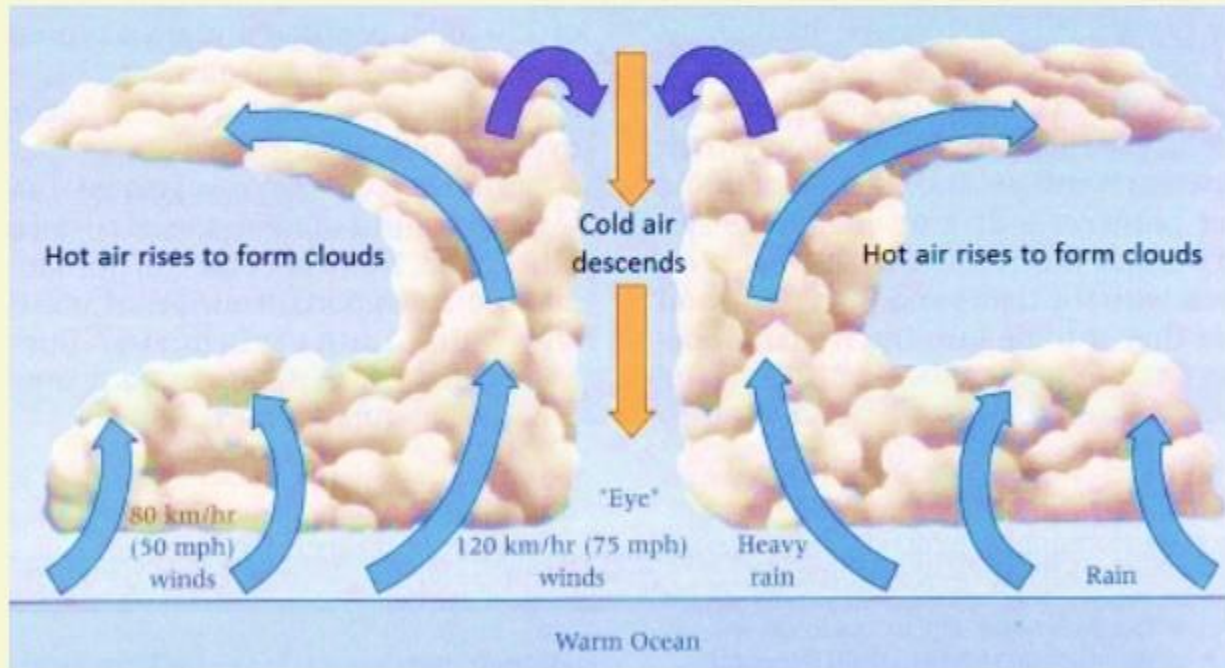
STRUCTURE OF TROPICAL CYCLONE



- 1** A mature tropical cyclone is characterised by the strong spirally circulating wind around the centre, called the eye.
- 2** The diameter of the circulating system can vary between 150 and 250km.
- 3** The eye is a region of calm with subsiding air.

TROPICAL CYCLONES

STRUCTURE OF TROPICAL CYCLONE



- 3** Around the eye is the **eye wall**, where there is a **strong spiralling ascent of air** to greater height reaching the tropopause.
- 4** The wind reaches maximum velocity in this region. Torrential rain occurs here
- 5** From the eye wall, **rain bands** may radiate and trains of cumulus and cumulonimbus clouds may drift into the outer region.

TROPICAL CYCLONES

STRUCTURE OF TROPICAL CYCLONE

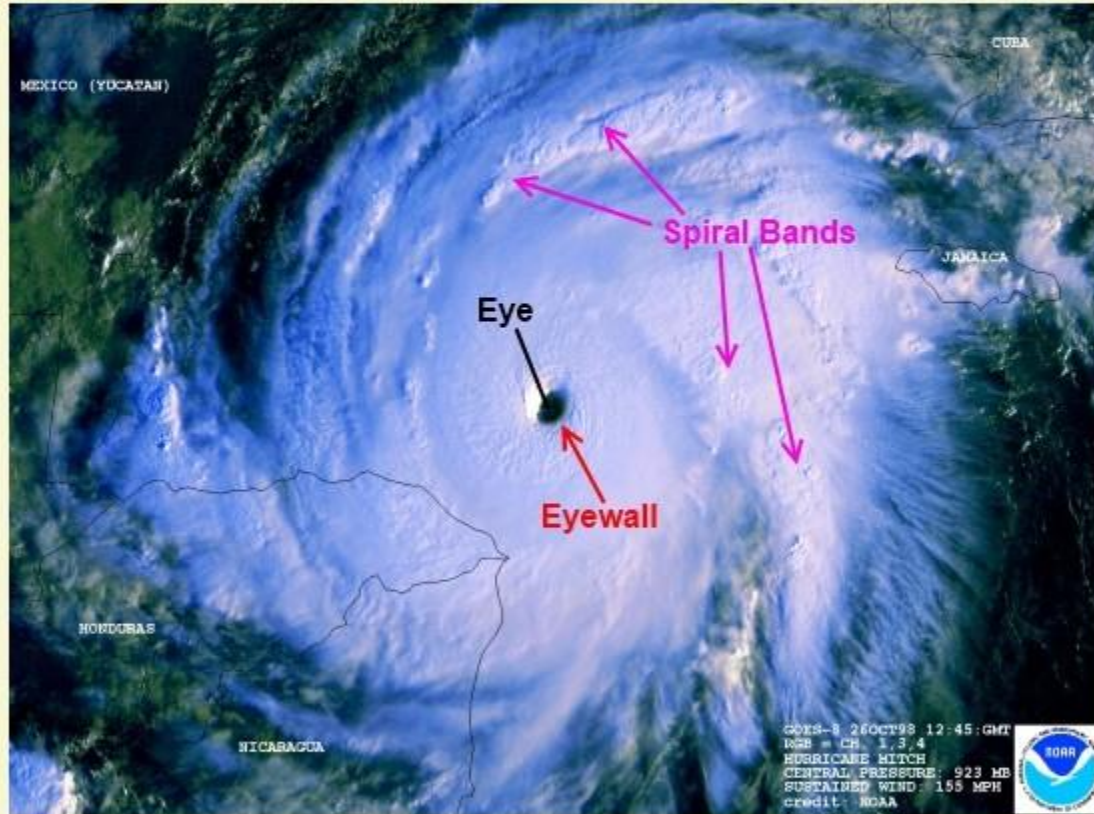


Image Courtesy: NOAA

6 The storm systems over the Bay of Bengal, Arabian sea and Indian ocean moves slowly about 300-500 km per day.

7 The cyclone creates storm surges and they inundate the costal low lands. The storm peters out on the land.

DIFFERENCE BETWEEN TROPICAL AND TEMPERATE CYCLONES

- 1** The temperate cyclone differs from the tropical cyclone in various ways. The temperate cyclones have a clear frontal system, which is not present in the tropical cyclones.
- 2** Temperate cyclones cover a larger area and can originate over the land and sea. Whereas tropical cyclones originate only over the seas and on reaching the land they dissipate.
- 3** The temperate cyclone affects a much larger area as compared to the tropical cyclone but the wind velocity in tropical cyclone is much higher and it is thus more destructive.
- 4** The temperate cyclones move from west to east but tropical cyclone, move from east to west.

S.No	TROPICAL CYCLONES	TEMPERATE CYCLONES
1	Smaller in size.	Larger in size and cover a large area.
2	Can originate only overseas and dissipate on reaching the land.	Can originate both over land and sea.
3	No frontal system found here.	Presence of frontal system with a cold and warm font.
4	Wind velocity is much higher attaining up to 400 Kmph are more destructive.	Wind velocity is less 18-32 Kmph.
5	Move from East to West as they are under the influence of trades.	Move from West to East as they are under the influence of westerlies.

TROPICAL CYCLONES

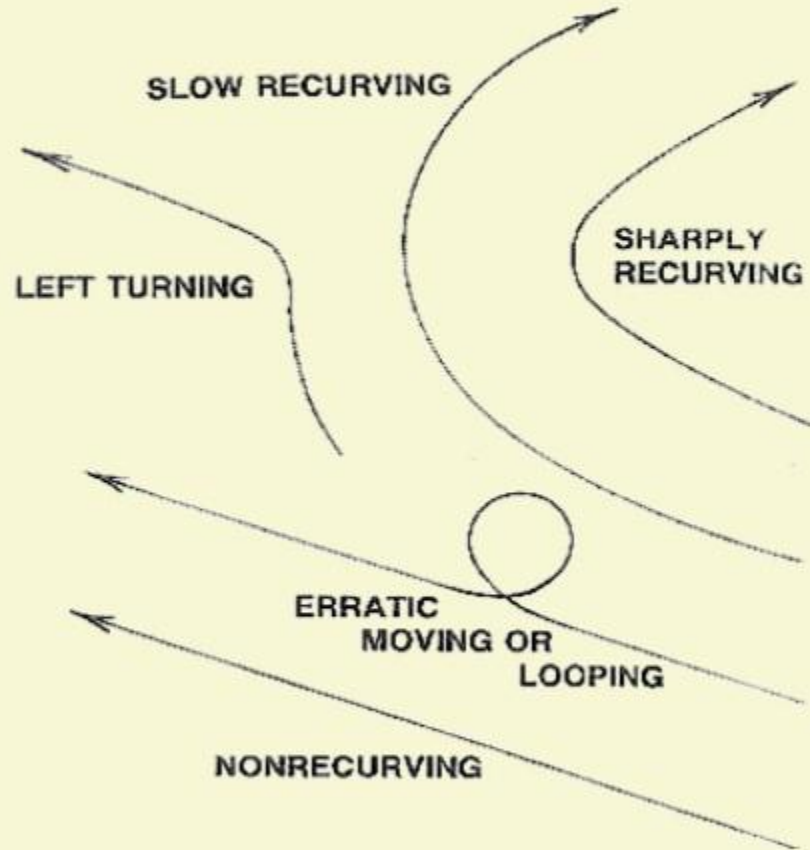
RECURVING CYCLONES

1 Tropical Cyclone (TC) Recurvature is the change in the cyclone's track from **westward to eastward and poleward**.

2 Recurving TC is defined as **poleward movement of a TC** owing to the change in the influence of the **wind systems** from the deep easterlies of the tropics into the strong upper tropospheric westerlies of the mid-latitudes.

3 Recurving TCs can be further classified with respect to how rapidly they recurve.

- 1. Sharply Recurving Cyclones -**
Sharply recurving cyclones are those which **change rapidly** from a westerward or northwestward course to a northeastward or a eastward course.
- 2. Slowly Recurving Cyclones -**
Slowly recurving cyclones are those which **change slowly** from a westerward or northwestward course to a northeastward or a eastward course.



REFERENCE

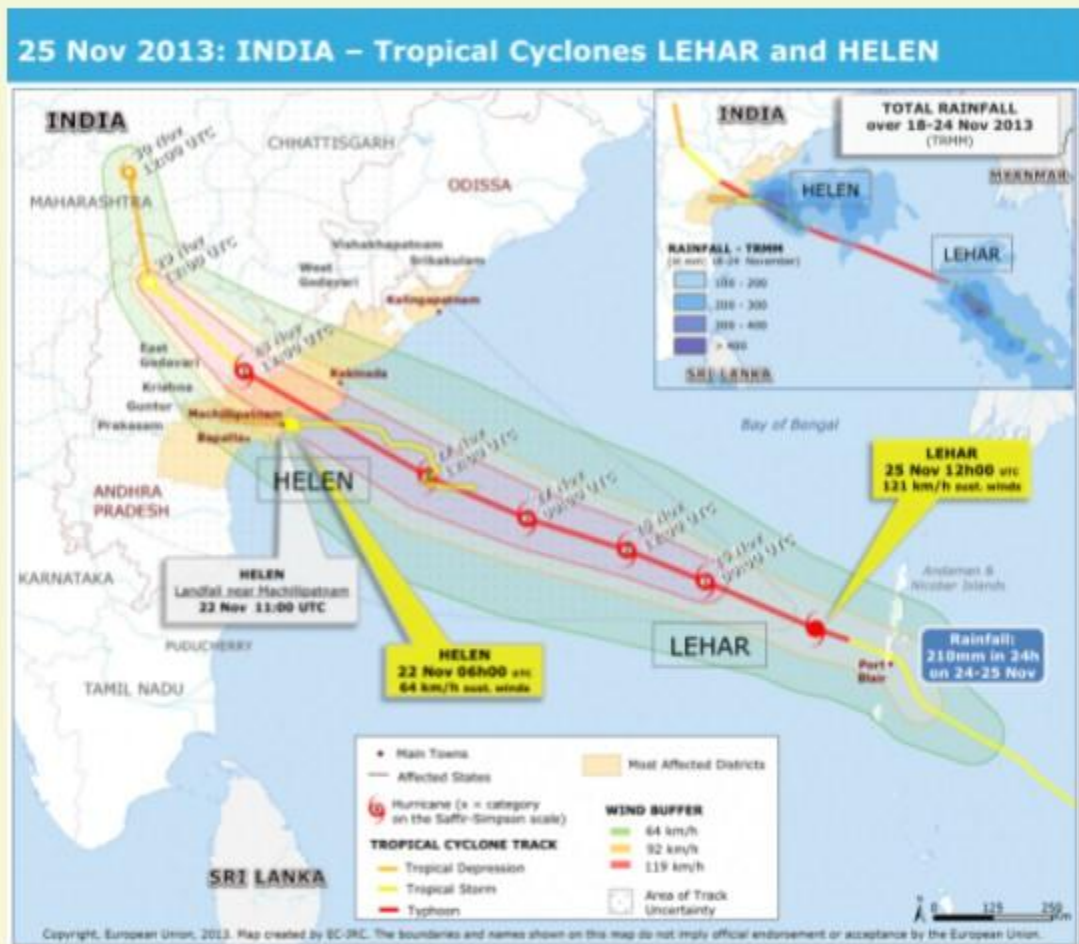
- *THE HINDU-‘RECURVING’ CYCLONES DRIED AUGUST RAINS-*
<http://www.thehindu.com/news/national/recurving-cyclones-dried-august-rains/article19694070.ece>

WHY WAS IT IN NEWS?

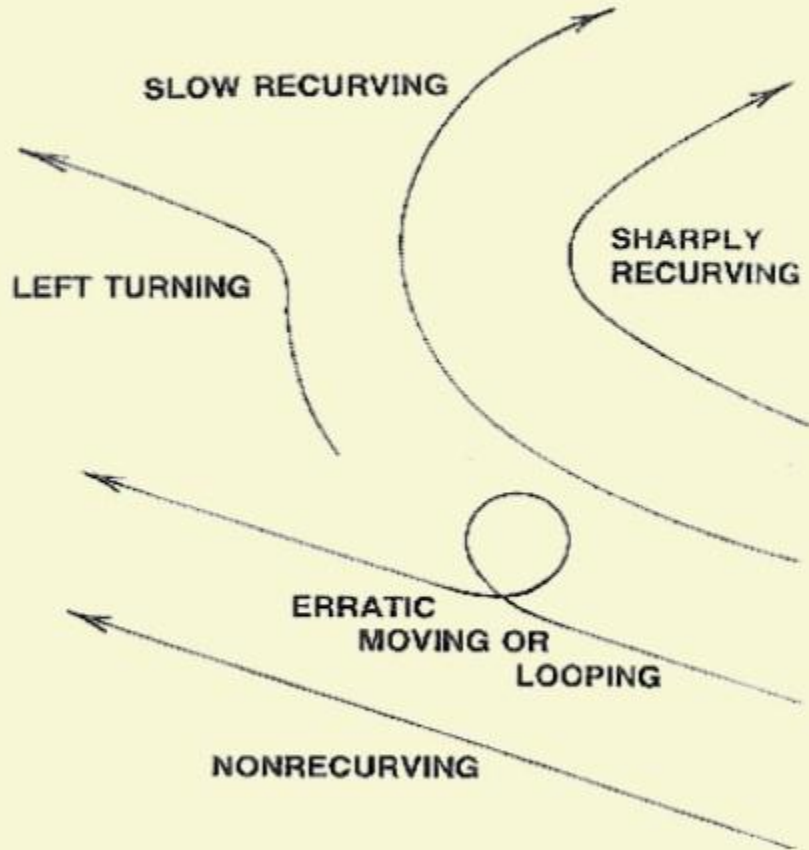
- Long-term data suggest that while there has been an increase in the number of tropical cyclones in India's neighbourhood there is **no clear trend** in re-curving ones.
- However, **these are back in the news** due to Cyclone Ockhi.



Path of Cyclone Ockhi



Path of Cyclone Lehar and Helen



REFERENCE

- *THE HINDU- WHAT IS THE LOWDOWN ON RE-CURVING CYCLONES?*

RECURVING CYCLONES

- It is found that the tendency of the cyclones to recurve is closely associated with the period of El-Nino.
- During El-Nino, when the trades(tropical easterlies) weaken, the cyclones nearer to the subtropical latitude have a tendency to recurve changing its direction towards the East/north-east rather than moving towards the west.

TROPICAL CYCLONES

RECURVING CYCLONES

- 1** During the monsoon months, cyclones in the Western Pacific move westwards towards India and aid rain-bearing systems over the sub-continent.
- 2** But during some years they 'recurve', or start to swing north-east, and do not give as much of a push to the rains as they do in the good monsoon years.
- 3** This re-curve frequently happens mostly during the El Nino years but this time(2017) it inexplicably occurred even when an El Nino hasn't yet taken shape.
- 4** El Nino years are those when sea surface temperatures in the east equatorial Pacific (near Peru) rise. This will weaken the trade winds and often dampen the monsoon .

QUESTION NO - 3

The term “Recurving cyclones” was recently seen in news. Recurving tropical cyclones in Western Pacific

1. Is a phenomenon that happens generally in El Niño years
2. Move toward the east with a poleward component

Which of the above are correct?

- | | |
|---------------|--------------------|
| a) 1 only | b) 2 only |
| c) Both 1 & 2 | d) Neither 1 nor 2 |

Answer: C

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